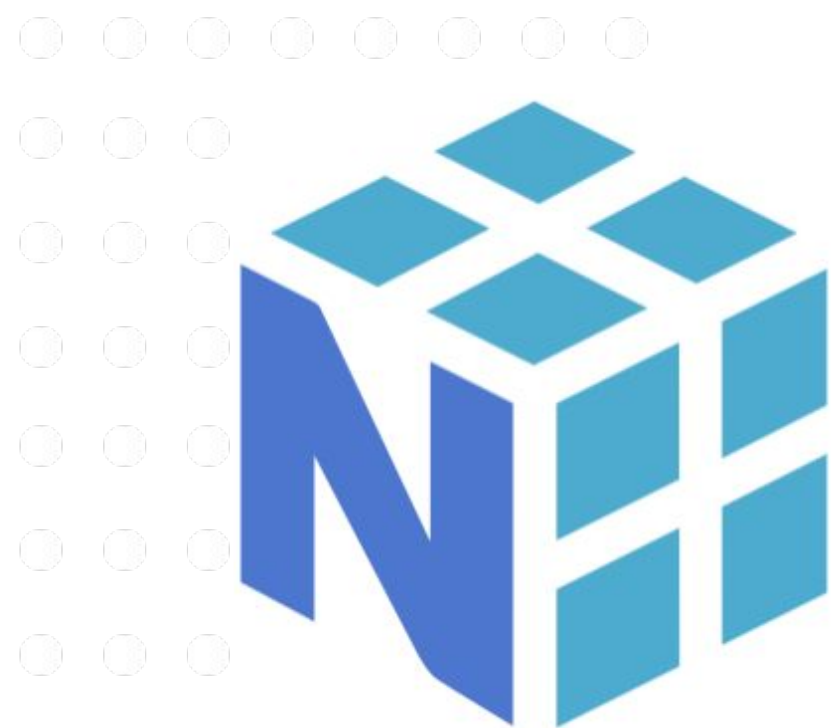


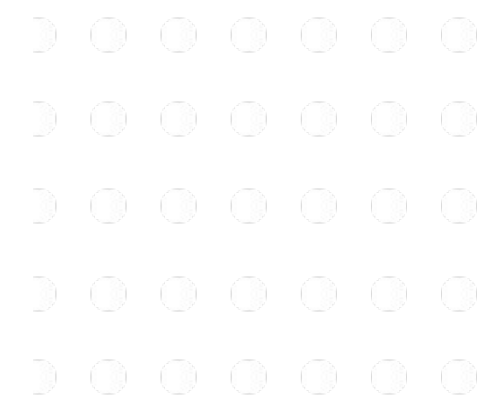


NUMPY

UPDATE

presented by
INESSA PAWSON (OPENTEAMS)

SCIPY'26 TOOLS PLENARY



FREE-THREADED PYTHON SUPPORT

- Resolved numerous scaling issues reported by the community
- NumPy 2.5 supports Python 3.15 (source level)
- 2.5.x release with 3.15 binaries is coming soon
- Support for new Python releases (3.14 and upcoming 3.15)

SIMD OPTIMIZATION

- Performance improvements across vectorized operations on modern CPUs
- Ongoing architecture-specific optimizations

TYPE ANNOTATIONS

- 100% type coverage, including numpy.ma
- Preliminary shape typing support introduced in NumPy 2.5
- Improved type annotations across numpy.linalg, numpy.fft, and dtype promotion logic
- Runtime-introspectable type signatures
- Fully typed ufuncs are underway for NumPy 2.6
- Better compatibility with mypy, Pyright, and other type checkers
- Downstream regression testing with mypy_primer

RELEASE INFRASTRUCTURE

- New numpy-release repository
- Improved release automation and software supply-chain security

SORTING INFRASTRUCTURE

NumPy 2.4:

- Modernized the sorting infrastructure with ArrayMethods
- Enables custom sorting for user-defined dtypes

NumPy 2.5: Added `descending=True` to `np.sort()` and `np.argsort()`

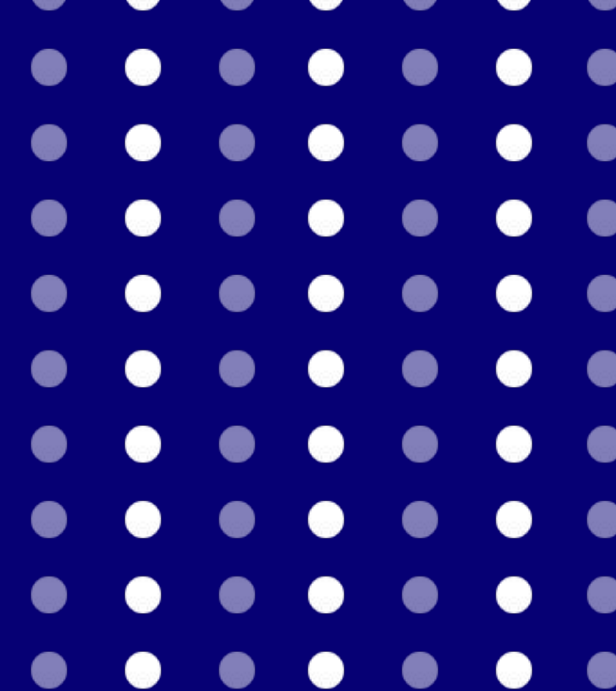
NumPy 2.6 (*coming soon*): Modernized partitioning and new `np.top_k()` function

SAFE TYPE CASTING

NumPy 2.4: New `casting="same_value"` mode

AI CONTRIBUTION POLICY

CONNECT WITH NUMPY TEAM



mailing list
or GitHub

NEWCOMERS' HOURS
monthly

COMMUNITY CALLS
biweekly

OPTIMIZATION TEAM MEETINGS
every third Monday

TRIAGE CALLS
biweekly